

Requirement Document for E-Library System

1. Introduction

The purpose of this document is to outline the requirements for an E-Library system that provides various functionalities to users, including user management, resource management, secure access, and personalized features like reading history and a favourite book list.

2. User Management

2.1 User Login

- Users should be able to log in using their email and password.

2.3 User Profile Management

- Users can view and edit their profile information such as Name, Email, and Password.
- Users should have the ability to update their password securely.

2.4 User Roles

- The system should support different user roles:
 - **Student:** Can read books, and mark as favourite, user can read maximum 20 books a month.
 - **Admin:** Has all permissions including managing user roles, overseeing the overall system, and generating reports. Can manage resources (add, update, delete), view user activity, and manage user access.
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3. Resource Management (Cataloging)

3.1 Add Resources

- Admin should be able to add new resources (books, articles, etc.) to the system.
- Each resource should have essential metadata such as Title, Author, Publisher, Publication Year, ISBN, and Description.
- Digital resources such as PDFs can be attached to the resources.

3.2 Update Resources

- Admin can update the metadata and file(s) associated with an existing resource.

3.3 Delete Resources

- Admin should be able to remove any digital resource from the library catalog.
- The system should confirm the deletion to avoid accidental loss of resources.

3.4 Categorization

- Resources should be categorized by subjects, genres, or themes for easier browsing.
- Admin can assign categories to each resource upon addition and update these categories as needed.

3.5 Resource Availability

- Users should only be able to access resources based on their roles (for example, some resources may only be available to students, others to admins).
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4. Search and Browse

4.1 Keyword Search

- Users should be able to search for resources by entering keywords.
- The system should support searching by resource title, author, and description.

4.2 Advanced Filters

- The system should allow users to filter search results by:
 - Author
 - Category/Genre
 - Publication Year
 - Resource Type (e.g., book, article, journal)

4.3 Browse by Categories/Subjects

- Users should be able to browse resources by different categories or subjects.
 - Categories should be clearly displayed in a menu or dropdown list for ease of navigation.
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5. Access Control & Security

5.1 User Roles and Permissions

- **Student:** Can view, borrow, mark books as read, rate books, and add books to their favourites list.
- **Admin:** Has all student privileges and can add, update, and delete resources, as well as view user activity.

5.2 Secure Access

- All sensitive data (such as passwords) should be securely encrypted.
- Access to certain resources should be role-based, ensuring that only authorized users can access specific materials.
- Secure login via HTTPS and password hashing techniques to protect user credentials.

5.3 Activity Logging

- The system should maintain an activity log to track user actions, such as resource additions, modifications, and deletions, especially for admin roles.
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6. Reading History/User Library

6.1 Mark as "Read"

- Logged-in users can mark a resource (book/article) as "read."
- Users should be able to track their progress and mark books as "completed."

6.2 View Reading History

- A dedicated section in the user's profile where they can view a list of all the books they have read.
- The list should include basic information about the book (title, author, date read).

6.3 Add Notes, Ratings, and Reviews

- Users should be able to add personal notes to the resources they've read.
 - Users can rate resources using a rating scale (e.g., 1 to 5 stars).
 - Users can also write reviews or comments on the resources they've read, which can be publicly visible (if the system allows).
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7. Favourite Book List

7.1 Mark as Favourite

- Users should be able to mark books as their favourite.
- The favourites list should be easily accessible and separate from the user's reading history.

7.2 Manage Favourites

- Users can view, remove, and manage their favourite book list.
- Users can add a book to their favourites from search results or resource pages.
